

Machine Learning Applications in Financial Leasing

Group Id - 25-26J-179

B.Sc. (Hons) Degree in Information Technology Specialization in
Information Technology

Department Of Information Technology
Sri Lanka Institute of Information Technology
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1 DECLARATION

I declare that this is my own work, and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology the non-exclusive right to produce and distribute my dissertation in whole or part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as article or books).

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2 ABSTRACT

This study presents the design and development of an integrated system that applies machine learning techniques to enhance operational efficiency, risk assessment, and customer service within the financial leasing sector. The system consists of multiple intelligent components, including an AI-driven conversational assistant, branch performance analysis module, default risk prediction model, and impairment analysis framework.

The AI-driven conversational assistant is developed as a multi-modal chatbot to provide real-time customer support for leasing companies. It utilizes a knowledge base integrated with Large Language Models (LLMs) through a Retrieval-Augmented Generation (RAG) approach, ensuring accurate and up-to-date responses. The system supports text, voice, and image-based interactions, improving accessibility and user experience while reducing dependency on human agents.

The branch performance analysis component leverages machine learning models such as Random Forest[1], XGBoost, and CatBoost to evaluate financial risk and employee performance at the branch level. By combining facility-level and employee-level data, the system generates predictive insights and supports both single and batch analysis, enabling data-driven decision-making.

The default risk prediction component employs a stacking ensemble model to classify customers into high, medium, and low-risk categories based on financial and behavioral data. With high prediction accuracy, this model assists in identifying potential defaulters and improving credit risk management.

Additionally, the impairment analysis component predicts loan impairment and Expected Credit Loss (ECL) using advanced feature engineering and ensemble learning techniques. The results demonstrate high accuracy in impairment prediction and provide valuable support for regulatory compliance and financial planning.

Overall, the proposed system demonstrates the effectiveness of machine learning applications in financial leasing by improving customer service, enhancing risk prediction, and supporting strategic decision-making. The integration of multiple intelligent components offers a scalable and efficient solution for modern leasing operations.

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6 LIST OF ABBREVIATIONS

Table 1 - Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
API	Application Programming Interface
PD	Probability of Default
ECL	Expected Credit Loss
IFRS	International Financial Reporting Standards
STT	Speech-to-Text
TTS	Text-to-Speech
DB	Database
UI	User Interface

7 INTRODUCTION

Financial leasing is a vital component of the modern financial industry, providing individuals and businesses with access to assets such as vehicles and equipment without requiring full upfront payment. Leasing companies play a key role in supporting economic activities by offering flexible financing solutions. However, with the increasing volume of customers and transactions, leasing institutions face significant challenges related to customer service, risk management, and operational efficiency.

Traditional leasing systems rely heavily on manual processes and rule-based decision-making, which can be time-consuming, error-prone and inefficient when handling large-scale data. In particular, assessing customer creditworthiness, predicting loan defaults, evaluating branch performance and providing timely customer support are critical areas that require improvement.

With the advancement of technology, Machine Learning (ML) has emerged as a powerful tool for solving complex problems in the financial sector. Machine learning enables systems to learn patterns from historical data and make accurate predictions or decisions without explicit programming. In the context of financial leasing, ML can be applied to automate decision-making processes, enhance risk assessment, and improve customer engagement.

This research focuses on the application of machine learning techniques in financial leasing through the development of an integrated system. The system includes multiple components such as an AI-driven conversational assistant, default risk prediction model, branch performance analysis, and impairment analysis. These components collectively aim to improve efficiency, accuracy, and service quality within leasing operations.

7.1 Background Literature

The adoption of machine learning in the financial sector has grown rapidly over the past decade, particularly in areas such as credit risk assessment, fraud detection, and customer relationship management. Previous studies have demonstrated that machine

learning models, including decision trees, random forests, and gradient boosting techniques, can significantly outperform traditional statistical methods in predicting financial risks [1].

In the leasing industry, credit risk prediction is one of the most critical applications. Research shows that ensemble learning methods, which combine multiple machine learning models, provide higher accuracy and robustness in predicting loan defaults compared to single-model approaches, particularly with advanced boosting algorithms such as XGBoost [2].

Additionally, the use of Artificial Intelligence in customer service has gained popularity through chatbot systems. AI-powered conversational agents are capable of handling large volumes of customer inquiries in real time, reducing operational costs and improving customer satisfaction. Recent advancements in Large Language Models (LLMs), especially transformer-based architectures, have significantly enhanced chatbot capabilities [3].

Furthermore, Retrieval-Augmented Generation (RAG) techniques improve response accuracy by combining language models with external knowledge sources, enabling context-aware and up-to-date responses [4].

Performance analysis in financial institutions has also evolved with the integration of machine learning. Combining financial data with employee performance metrics provides deeper insights into organizational efficiency and supports predictive decision-making [5].

Impairment and Expected Credit Loss (ECL) prediction have become increasingly important with the introduction of financial reporting standards such as IFRS 9, which require forward-looking credit risk assessment models [6].

Overall, existing literature highlights the significant potential of machine learning in transforming financial leasing operations. However, there is a need for integrated systems that combine multiple ML applications into a unified solution. This research addresses that gap by developing a comprehensive system that incorporates customer support, risk prediction, and performance analysis within a single framework.

7.2 Research Gap

Although machine learning has been widely applied in the financial sector, particularly in areas such as credit risk prediction, fraud detection, and customer service automation, its application within the financial leasing industry remains relatively limited and fragmented. Most existing studies focus on single problem domains, such as loan default prediction or chatbot-based customer support, rather than providing a comprehensive, integrated solution tailored specifically for leasing operations.

In the context of financial leasing, several key challenges are not fully addressed by current research. Firstly, many traditional risk assessment models rely on statistical techniques or isolated machine learning models, which may not capture the complex relationships between financial, behavioral, and operational data. There is a lack of advanced ensemble-based approaches that combine multiple models to improve prediction accuracy and reliability.

Secondly, while AI-powered chatbots are increasingly used in customer service, many systems depend heavily on pre-trained language models without incorporating dynamic, company-specific knowledge bases. This often results in generic or outdated responses, reducing the effectiveness of such systems in real-world leasing environments.

Additionally, existing research rarely integrates multiple functional areas—such as customer interaction, default risk prediction, branch performance evaluation, and impairment analysis into a single unified framework. As a result, decision-making processes remain siloed, limiting the ability of organizations to gain a holistic understanding of their operations.

Furthermore, there is limited research on applying machine learning to support regulatory requirements such as Expected Credit Loss (ECL) estimation under IFRS 9 within leasing companies. Many existing approaches do not fully utilize advanced feature engineering and ensemble learning techniques to enhance prediction performance.

Therefore, there is a clear need for an integrated, machine learning-based system that combines multiple analytical and operational components into a single platform. This

research aims to address these gaps by developing a comprehensive solution that enhances customer service, improves risk prediction accuracy, and supports data-driven decision-making in financial leasing.

7.3 Research Problem

Financial leasing companies face several operational and analytical challenges due to increasing customer demand, large volumes of financial data, and the need for accurate risk assessment. Traditional systems rely heavily on manual processes and basic statistical methods, which are often insufficient for handling complex and dynamic financial data.

One of the major problems is the difficulty in accurately predicting customer default risk using conventional methods. These approaches may fail to capture hidden patterns in customer financial behavior, leading to poor credit decisions and increased financial losses. Additionally, leasing companies struggle to evaluate branch performance effectively, as existing systems do not integrate both financial and employee-related factors into a unified analysis.

Another significant issue is inefficient customer service. Many leasing companies still depend on human agents to handle customer inquiries, resulting in delays, increased operational costs, and inconsistent service quality. Existing chatbot solutions often provide generic responses and lack integration with real-time company data.

Furthermore, impairment and Expected Credit Loss (ECL) estimation, which are critical for financial reporting and regulatory compliance, remain complex and time-consuming tasks when performed using traditional techniques.

Therefore, the main research problem is the lack of an integrated, intelligent system that can effectively combine customer service automation, risk prediction, performance analysis, and financial forecasting within the financial leasing domain.

7.4 Research Objectives

The primary objective of this research is to develop a comprehensive and integrated system that leverages machine learning techniques to improve key operational areas within the financial leasing industry. This study aims to enhance the efficiency, accuracy, and reliability of leasing processes by incorporating intelligent automation, predictive analytics, and real-time decision support mechanisms.

Specifically, the research focuses on addressing major challenges such as inefficient customer service, inaccurate risk assessment, lack of performance evaluation tools, and difficulties in financial forecasting. By applying advanced machine learning models and modern AI technologies, the proposed system seeks to provide a scalable and data-driven solution that supports both operational and strategic decision-making in leasing companies.

1.4.1 Specific Objectives

Sub-Objective 01: Development of AI-Driven Conversational Assistant

- To design and implement an intelligent chatbot system for leasing customer support.
- To integrate a knowledge base containing company-specific information for accurate responses.
- To apply Retrieval-Augmented Generation (RAG) for context-aware answer generation.
- To enable multi-modal interaction including text, voice, and image-based inputs.
- To implement a human agent handoff mechanism for complex queries.

Sub-Objective 02: Development of Default Risk Prediction Model

- To collect and preprocess customer financial and behavioral data.
- To apply feature engineering techniques to improve prediction performance.
- To develop machine learning models such as Random Forest[1], XGBoost, and CatBoost.
- To implement a stacking ensemble model for improved accuracy.

- To classify customers into high, medium, and low-risk categories based on probability of default.

Sub-Objective 03: Development of Branch Performance Analysis Component

- To analyze facility-level financial data and employee performance data.
- To develop predictive models for evaluating branch efficiency.
- To implement hybrid ensemble techniques for improved prediction reliability.
- To provide both single and batch analysis functionalities.
- To generate visual insights such as confusion matrices and feature importance charts.

Sub-Objective 04: Development of Impairment Analysis Model

- To perform advanced feature engineering for financial risk analysis.
- To develop machine learning models for predicting impairment and Expected Credit Loss (ECL).
- To implement ensemble learning techniques for improved accuracy.
- To evaluate model performance using metrics such as R^2 , RMSE, and MAE.
- To support financial reporting and regulatory compliance (e.g., IFRS 9).

Sub-Objective 05: System Integration and Deployment

- To integrate all components into a unified system architecture.
- To develop a FastAPI-based backend for real-time predictions.
- To ensure scalability and efficient data processing.
- To design a user-friendly interface for both customers and administrators.
- To test and validate the system for real-world application.

8 METHODOLOGY

8.1 Methodology

This research adopts a machine learning–driven approach to develop an integrated system for financial leasing applications, focusing on customer service automation, risk prediction, performance evaluation, and financial loss estimation. The methodology combines data-driven modeling, system integration, and real-time processing to ensure accuracy, scalability, and practical usability.

The overall process begins with **data collection and preprocessing**. Multiple datasets related to leasing operations were utilized, including customer financial data, loan facility details, and employee performance data. These datasets contain attributes such as facility amount, interest rates, repayment history, arrears, employee characteristics, and operational indicators. Data preprocessing techniques such as data cleaning, handling missing values, encoding categorical variables, and normalization were applied to ensure data quality and compatibility with machine learning models.

Following preprocessing, **feature engineering** was performed to enhance model performance. New features were derived from existing variables to capture meaningful relationships within the data. These include ratio-based features, risk indicators, and interaction terms that help models better understand financial behavior and risk patterns.

The system consists of four main components, each developed using appropriate methodologies:

First, the **AI-driven conversational assistant** was developed using a Retrieval-Augmented Generation (RAG) approach. A structured knowledge base containing verified company information is stored in a database and converted into vector embeddings. When a user query is received, semantic similarity search is performed to retrieve relevant information. This information is then passed to a Large Language Model (LLM) to generate accurate and context-aware responses. The system supports text, voice, and image-based inputs, ensuring accessibility and improved user interaction.

Second, the **default risk prediction component** was implemented using a stacking ensemble model. Multiple machine learning algorithms, including Random Forest[1], XGBoost, and CatBoost, were trained on financial and behavioral data. Their outputs were combined to produce a final prediction with high accuracy. The model generates a probability of default, which is categorized into risk levels (high, medium, and low), enabling effective credit risk management.

Third, the **branch performance analysis component** evaluates both financial risk and employee performance. Various models such as Decision Trees, Random Forest[1], and Gradient Boosting techniques were used. A hybrid ensemble method was applied to combine predictions and improve reliability. The system supports both single and batch analysis modes and provides visualizations such as confusion matrices and feature importance graphs to support decision-making.

Finally, the **impairment analysis component** predicts loan impairment and Expected Credit Loss (ECL). Multiple advanced machine learning models were trained and combined using a stacking ensemble approach. Model performance was evaluated using metrics such as R^2 score, RMSE, and MAE to ensure accuracy and generalization. The system is implemented using a **FastAPI-based backend**, enabling real-time prediction and seamless communication between components. Overall, this methodology ensures a robust, scalable, and efficient system for applying machine learning in financial leasing.

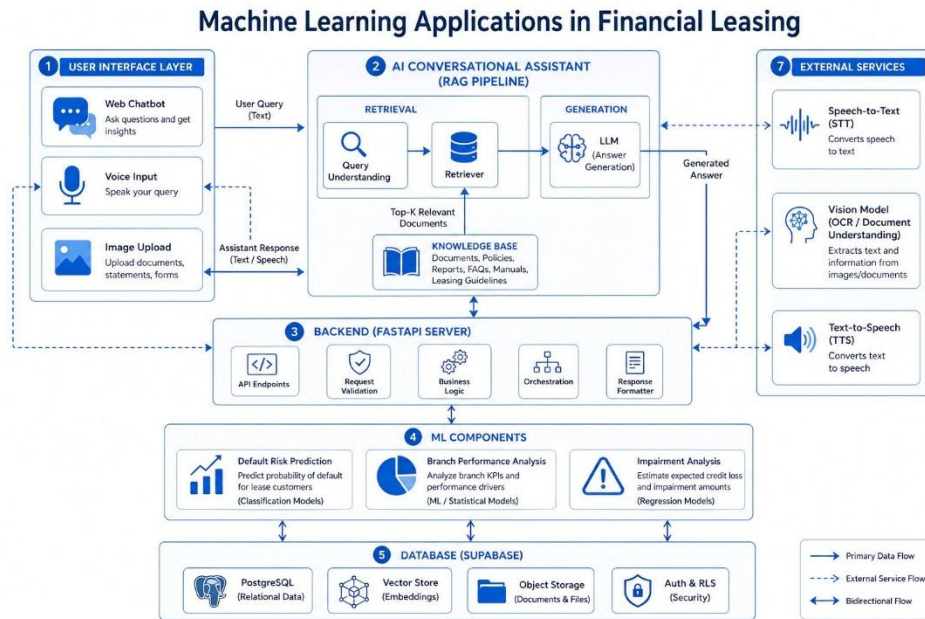


Figure 1- System Overview

8.2 Commercialization aspects of the product

The proposed system, Machine Learning Applications in Financial Leasing, demonstrates strong commercialization potential due to its ability to solve critical challenges faced by modern leasing companies. These challenges include inefficient customer service, inaccurate risk assessment, lack of integrated performance monitoring, and complex financial forecasting processes. By combining multiple machine learning based components into a unified platform, the system offers a comprehensive, scalable, and cost-effective solution suitable for real-world industry adoption.

One of the key strengths of the product is its **modular and flexible architecture**. Each component of the system such as the AI-driven conversational assistant, default risk prediction module, branch performance analysis, and impairment analysis—can function independently or as part of an integrated system. This allows organizations to adopt specific modules based on their operational needs and gradually scale up to the full system. Such flexibility makes the product attractive to both small-scale leasing companies and large financial institutions.

The **AI-driven conversational assistant** provides significant commercial value by automating routine customer interactions. In traditional systems, customer service heavily depends on human agents, leading to increased labor costs and delays in response times. By implementing an intelligent chatbot that delivers real-time, accurate responses, companies can significantly reduce operational expenses while improving customer satisfaction. The inclusion of multi-modal capabilities—text, voice, and image enhances accessibility and provides a modern, user-friendly experience, increasing the product’s market competitiveness.

The **default risk prediction component** contributes directly to financial performance by enabling better credit decision-making. By accurately classifying customers into different risk categories, leasing companies can reduce the likelihood of loan defaults and minimize financial losses. This predictive capability is especially valuable in competitive markets where efficient risk management is essential for long-term sustainability.

Similarly, the **branch performance analysis module** provides actionable insights into both financial operations and employee productivity. By identifying underperforming branches or high-risk facilities, organizations can take proactive measures to improve efficiency. This leads to better resource allocation, improved employee management, and enhanced overall performance. The ability to visualize data and generate real-time insights further increases the practical value of this component.

Another important commercialization aspect is the **impairment analysis and Expected Credit Loss (ECL) prediction** feature. This component supports financial institutions in complying with regulatory standards such as IFRS 9. By automating complex financial calculations and providing accurate forecasts, the system reduces manual workload and ensures compliance with regulatory requirements. This makes the product particularly appealing to organizations that must adhere to strict financial reporting standards.

From a technological perspective, the system is designed using **modern, scalable, and cost-efficient technologies**. The use of cloud-based infrastructure allows the system to handle large volumes of data and user interactions without performance

issues. The FastAPI backend ensures real-time processing and smooth communication between system components. Additionally, the reliance on open-source tools and APIs reduces development and maintenance costs, making the product economically viable for commercialization.

The inclusion of a **user-friendly admin interface** is another significant advantage. Non-technical staff can easily manage the system, update the knowledge base, and monitor performance without requiring advanced technical expertise. This reduces dependency on specialized IT personnel and lowers operational complexity, which is a critical factor for successful adoption in business environments.

In terms of business models, the system can be offered as a **Software-as-a-Service (SaaS)** solution. This allows companies to subscribe to the platform and access its features through the cloud, eliminating the need for expensive infrastructure setup. The SaaS model also provides opportunities for continuous updates, maintenance, and customer support, ensuring long-term customer engagement and recurring revenue generation. Additionally, customization services can be offered to tailor the system to specific organizational requirements, further increasing its market value.

Furthermore, the system has the potential to be expanded beyond leasing companies to other financial sectors such as banking, insurance, and microfinance. This scalability broadens its market reach and enhances its commercial viability.

In conclusion, the proposed system offers a highly marketable solution by combining advanced machine learning capabilities with practical business applications. Its flexibility, scalability, cost-effectiveness, and ability to improve operational efficiency make it a strong candidate for commercialization in the financial leasing industry and beyond.

9 Results & Discussion

This chapter presents the results obtained from the implementation of the proposed machine learning based system for financial leasing, along with a detailed discussion of the findings. The system consists of four main components: the AI-driven conversational assistant, default risk prediction model, branch performance analysis

module, and impairment analysis component. Each component was evaluated based on its performance, accuracy, and practical applicability.



Figure 2- Gantt Chat

9.1 Results

The results of the developed system demonstrate high effectiveness across all components.

The **AI-driven conversational assistant** successfully provided accurate and context-aware responses to user queries by utilizing a Retrieval-Augmented Generation (RAG) approach. The integration of a dynamic knowledge base ensured that responses remained up-to-date and relevant. The system effectively handled text, voice, and image inputs, providing a seamless user experience. The human agent handoff feature also functioned efficiently, ensuring uninterrupted customer support when required.

The **default risk prediction model** achieved very high accuracy through the use of a stacking ensemble approach. Individual models such as Random Forest[1], XGBoost, and CatBoost performed with accuracy levels above 99%, while the combined stacking model achieved the highest performance. The model successfully classified customers into high, medium, and low-risk categories based on their probability of default, demonstrating strong predictive capability.

The **branch performance analysis component** produced reliable results in evaluating both financial risk and employee performance. Machine learning models accurately classified facilities as “good” or “poor,” while the hybrid ensemble model provided stable and consistent employee performance scores. Visualization tools such as confusion matrices and feature importance graphs confirmed the effectiveness of the models and highlighted key influencing factors.

The **impairment analysis component** achieved excellent results in predicting loan impairment, with accuracy levels exceeding 99% for certain models. The prediction of Expected Credit Loss (ECL) also showed strong performance, with the stacking ensemble model achieving over 92% accuracy. These results indicate the system’s ability to support financial forecasting and regulatory compliance.

9.2 Research Findings

Several important findings emerged from the implementation and evaluation of the system.

Firstly, the use of **ensemble learning techniques** significantly improved prediction accuracy compared to individual models. Combining multiple algorithms allowed the system to capture complex patterns in financial data, resulting in more reliable predictions.

Secondly, the integration of a **knowledge-based chatbot with LLMs** proved highly effective in delivering accurate and real-time customer support. The use of a dynamic knowledge base ensured that responses were not only intelligent but also aligned with company-specific information.

Thirdly, the study found that **feature engineering plays a critical role** in improving model performance. The creation of additional features such as risk indicators, ratios, and interaction terms enhanced the predictive power of machine learning models.

Additionally, combining **financial and employee data** in branch performance analysis provided a more comprehensive understanding of operational efficiency. This approach enabled better identification of performance issues and risk factors.

Finally, the system demonstrated that machine learning can effectively support **regulatory requirements**, particularly in impairment and ECL prediction, reducing the complexity of financial reporting processes.

9.3 Discussion

The results obtained from the implementation of the proposed system highlight the significant potential of machine learning in transforming financial leasing operations. Each component of the system contributes to solving key challenges in the industry, including customer service inefficiencies, risk assessment limitations, and lack of integrated decision-making tools.

The **default risk prediction component** demonstrated exceptionally high accuracy, particularly through the use of a stacking ensemble model. By combining multiple algorithms such as Random Forest[1], XGBoost, and CatBoost, the system was able to capture complex relationships within customer financial and behavioral data. This confirms that ensemble learning techniques are more effective than single-model approaches in predicting loan default risk. The classification of customers into high, medium, and low-risk categories provides practical value for leasing companies, enabling them to make informed credit decisions and reduce potential financial losses.

Similarly, the **branch performance analysis component** proved to be highly effective in evaluating both financial and operational performance. By integrating facility-level data with employee-related metrics, the system offers a more comprehensive understanding of branch efficiency. The use of hybrid ensemble methods further improved prediction stability and reliability. Visualization tools such as confusion matrices and feature importance analysis provided valuable insights into the key factors influencing performance, allowing management to identify weaknesses and implement targeted improvements.

The **impairment analysis component** also delivered strong results, particularly in predicting loan impairment with very high accuracy. This demonstrates the effectiveness of advanced feature engineering and ensemble modeling in capturing financial risk patterns. Although the prediction accuracy for Expected Credit Loss

(ECL) was slightly lower compared to impairment prediction, it still provided meaningful insights for financial forecasting and regulatory compliance. This component is especially important for supporting standards such as IFRS 9, where accurate estimation of future losses is required.

The **AI-driven conversational assistant** significantly enhances customer service by providing real-time, accurate, and context-aware responses. The use of a knowledge base combined with Retrieval-Augmented Generation (RAG) ensures that responses are grounded in verified company data rather than generic information. The inclusion of multi-modal interaction (text, voice, and image) improves accessibility and user experience, making the system more practical for real-world use. Additionally, the human agent handoff feature ensures that complex queries are handled effectively, maintaining a balance between automation and human support.

Despite these positive outcomes, certain limitations should be acknowledged. The performance of machine learning models heavily depends on the quality and completeness of the input data. Inaccurate or missing data can negatively impact prediction results. Furthermore, the system may require continuous updates and retraining to adapt to changing financial conditions and customer behavior patterns.

Overall, the discussion confirms that the proposed system successfully addresses key gaps in financial leasing by integrating multiple machine learning applications into a single platform. The combination of high predictive accuracy, real-time processing, and user-friendly design makes the system a valuable tool for improving efficiency, reducing risk, and supporting data-driven decision-making in leasing companies.

further

9.4 Summary of Each Student's contribution

- This research project was carried out collaboratively, with each team member contributing to different components of the system based on their area of focus and expertise.
- One member was responsible for the development of the **AI-driven conversational assistant**. This included designing the chatbot architecture,

integrating the knowledge base, implementing the Retrieval-Augmented Generation (RAG) approach, and enabling multi-modal interactions such as text, voice, and image processing. Additionally, this member contributed to implementing the human agent handoff feature and ensuring smooth user interaction.

- Another member focused on the **default risk prediction component**. Their contribution included data preprocessing, feature selection, and the development of machine learning models. They implemented a stacking ensemble approach using algorithms such as Random Forest[1], XGBoost, and CatBoost, and evaluated model performance to achieve high prediction accuracy. They also worked on risk classification logic and API integration.
- A third member developed the **branch performance analysis component**. This involved building predictive models to evaluate both facility-level risk and employee performance. The member also implemented hybrid ensemble techniques, designed visualization outputs such as confusion matrices and feature importance graphs, and ensured the system could support both single and batch analysis.
- Another member was responsible for the **impairment analysis component**. Their work included advanced feature engineering, model training using multiple machine learning algorithms, and implementing a stacking ensemble approach for predicting impairment and Expected Credit Loss (ECL). They also handled performance evaluation using multiple metrics and prepared the models for deployment.
- All team members collaboratively contributed to **system integration, testing, and documentation**. This included connecting frontend and backend components, ensuring real-time functionality using APIs, and preparing the final report. Regular discussions and coordination among team members ensured the successful completion of the project.

10 CONCLUSION

This research study explored the application of machine learning techniques in the financial leasing industry through the development of an integrated, intelligent system. The primary aim was to address key challenges faced by leasing companies, including inefficient customer service, limitations in risk assessment, lack of performance evaluation tools, and complexities in financial forecasting. The results demonstrate that machine learning can significantly enhance the efficiency, accuracy, and overall effectiveness of leasing operations.

One of the major contributions of this research is the development of an **AI-driven conversational assistant** that improves customer service by providing real-time, accurate, and context-aware responses. By integrating a knowledge base with advanced language models using a Retrieval-Augmented Generation approach, the system ensures that responses are both reliable and up-to-date. The inclusion of multi-modal interaction text, voice, and image further enhances accessibility and user experience. This component reduces the dependency on human agents while maintaining high service quality.

The **default risk prediction component** plays a crucial role in improving financial decision-making. By utilizing a stacking ensemble model that combines multiple machine learning algorithms, the system achieves high prediction accuracy. This enables leasing companies to classify customers based on their risk levels and take proactive measures to minimize potential losses. The ability to accurately predict loan defaults is a significant advantage in managing financial risk.

The **branch performance analysis component** provides valuable insights into both financial operations and employee productivity. By combining facility-level and employee-level data, the system offers a comprehensive evaluation of branch efficiency. The use of hybrid ensemble techniques improves prediction reliability, while visualization tools support better understanding and decision-making. This component helps organizations identify weaknesses and optimize performance.

In addition, the **impairment analysis component** supports financial planning and regulatory compliance by predicting loan impairment and Expected Credit Loss

(ECL). The use of advanced feature engineering and ensemble models results in high prediction accuracy, particularly for impairment estimation. This component is especially important for meeting financial reporting standards and improving long-term financial stability.

Despite the strong performance of the system, some limitations exist. The effectiveness of machine learning models depends heavily on the quality and availability of data. Incomplete or inconsistent data may affect prediction accuracy. Furthermore, continuous monitoring and updating of models are required to ensure long-term reliability, especially in dynamic financial environments.

Overall, this research successfully demonstrates that integrating multiple machine learning applications into a single system can significantly improve the performance of financial leasing operations. The proposed solution provides a scalable, efficient, and cost-effective approach to enhancing customer service, risk management, and decision-making. It offers strong potential for real-world implementation and future expansion into other areas of the financial sector.

11 Glossary

- **Artificial Intelligence (AI)**

The simulation of human intelligence in machines that are programmed to think, learn, and make decisions.

- **Machine Learning (ML)**

A subset of artificial intelligence that enables systems to learn from data and improve their performance without being explicitly programmed.

- **Large Language Model (LLM)**

An advanced AI model trained on large amounts of text data to understand and generate human-like language.

- **Retrieval-Augmented Generation (RAG)**

A technique that combines information retrieval from a knowledge base with language generation to produce accurate and context-aware responses.

- **Knowledge Base**
A structured collection of information used by systems to retrieve and provide accurate answers.
- **Vector Embedding**
A numerical representation of data (such as text) that captures its meaning and allows similarity comparison.
- **Default Risk**
The likelihood that a borrower will fail to repay a loan according to the agreed terms.
- **Probability of Default (PD)**
A metric used to estimate the probability that a borrower will default on a loan.
- **Ensemble Learning**
A machine learning technique that combines multiple models to improve prediction accuracy.
- **Stacking Ensemble Model**
An advanced ensemble method that combines predictions from multiple models using a meta-model.
- **Random Forest**
A machine learning algorithm that uses multiple decision trees to improve prediction accuracy.
- **XGBoost**
An optimized gradient boosting algorithm known for high performance and efficiency.
- **CatBoost**
A gradient boosting algorithm that handles categorical data effectively.
- **Feature Engineering**
The process of creating new input variables from raw data to improve model performance.
- **Branch Performance Analysis**
The evaluation of financial and operational performance at the branch level within an organization.

- **Expected Credit Loss (ECL)**
A financial metric that estimates potential future losses from loans, required under accounting standards such as IFRS 9.
- **Impairment**
The reduction in the value of a financial asset when it is expected that the full amount will not be recovered.
- **FastAPI**
A modern web framework used to build APIs for machine learning models and applications.
- **API (Application Programming Interface)**
A set of rules that allows different software systems to communicate with each other.
- **Multi-modal System**
A system that supports multiple types of input such as text, voice, and images.
- **Speech-to-Text (STT)**
Technology that converts spoken language into written text.
- **Text-to-Speech (TTS)**
Technology that converts written text into spoken audio.

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13 APPENDICES